

Contraction mapping and Banach fixed point theorem

Let (X, d) be a metric space. A mapping $T: X \rightarrow X$ is said to be a contraction mapping on X if $\exists$ a real no. α with $0 < \alpha < 1$ such that $d(Tx, Ty) \leq \alpha d(x, y) \forall x, y \in X$.
The contraction mapping is uniformly continuous on X .

Ex: - ① Let $f: [a, b] \rightarrow [a, b]$ be a differentiable funcⁿ with $|f'(x)| \leq k \forall x \in [a, b]$ & $k < 1$. Then f is a contraction mapping on $[a, b]$.

② $T(x) = x^n$ where $0 \leq x \leq \frac{1}{n+1}$, $n \geq 1$ then T is a contraction mapping on $[0, \frac{1}{n+1}]$

③ Let $f: [0, 1] \rightarrow [0, 1]$ defined by $f(x) = \frac{1}{1+x}$. Is f a contraction on $[0, 1]$?

$$\Rightarrow d(Ty, Tx)$$

$$= \left| \frac{1}{1+y} - \frac{1}{1+x} \right| \leq \left| \frac{y-x}{(1+y)(1+x)} \right|$$
$$\leq \left| \frac{y}{1+y} \right| + \left| \frac{x}{1+x} \right|$$

$$0 < y < 1$$

$$\Rightarrow 1 < 1+y < 2$$

$$\Rightarrow \frac{1}{2} < \frac{1}{1+y} < 1$$

$$< |y - x|$$

$$\alpha = 1$$

$\therefore$ Not contraction mapping.

Fixed point

A point $x \in X$ is called fixed point of X of the mapping T if $T(x) = x$. A mapping has no or one or more than one fixed point.

20/3 Banach fixed point or contraction mapping principle.

Let (X, d) be a complete metric space and $T: X \rightarrow X$ be a contraction mapping then T has a unique fixed pt in X .

$\Rightarrow$ let $x_0 \in X$ and $\{x_n\}_{n \geq 0}$ be the sequence defined iteratively by $x_{n+1} = T(x_n)$, $n \geq 0$. We have to prove that $\{x_n\}$ is Cauchy sequence.

For $p = 1, 2, 3, \dots$ we have $d(x_{p+1}, x_p) \stackrel{④}{=} d(Tx_p, Tx_{p-1}) \leq \alpha d(x_p, x_{p-1})$

Where $0 < \alpha < 1$ such that $d(Tx, Ty) \leq \alpha d(x, y) \forall x, y \in X$. $\xrightarrow{\text{①}}$

Now applying ① repeatedly we can write

$$d(x_{p+1}, x_p) \leq \alpha d(x_p, x_{p+1}) \leq \alpha^2 d(x_{p-1}, x_p) \leq \dots \leq \alpha^p d(x_1, x_0).$$

let m, n be two +ve integer such that $m \geq n \geq 1$.

Then by triangular inequality

$$d(x_m, x_n) \leq d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \dots + d(x_{n+1}, x_n)$$

$$\leq (\alpha^{m-1} + \alpha^{m-2} + \dots + \alpha^n) d(x_1, x_0)$$

$$= \alpha^n (\alpha^{m-n-1} + \alpha^{m-n-2} + \dots + 1) d(x_1, x_0)$$

$$= \alpha^n \left(\sum_{k=0}^{m-n-1} \alpha^k \right) d(x_1, x_0)$$

$$\leq \alpha^n \left(\sum_{k=0}^{\infty} \alpha^k \right) d(x_1, x_0) = \frac{\alpha^n}{1-\alpha} d(x_1, x_0)$$

But $\lim_{n \rightarrow \infty} \alpha^n = 0$.

This shows that $\{x_n\}$ is a Cauchy sequence in X .

Since X is complete $\{x_n\}$ converges to some limit say $x^* \in X$ i.e. $\lim_{n \rightarrow \infty} x_n = x^*$. We have to show that $Tx^* = x^*$.

Since T being a contraction mapping ~~on~~ then T is uniformly continuous on X and hence it is continuous on X .

$$\text{And therefore } Tx^* = T\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} T(x_n) \\ = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

$\Rightarrow x^*$ is a fixed pt.

We now show that x^* is unique.

Let $y^* \in X$ is another fixed pt of T then

$$0 \leq d(x^*, y^*) = d(Tx^*, Ty^*) \leq \alpha d(x^*, y^*) < d(x^*, y^*)$$

$$\therefore d(x^*, y^*) = 0$$

$$\Rightarrow x^* = y^*$$

Ex-①

$T: \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = x+1$ in other words there are many examples of operators with fixed point that are not contraction mapping.

$$T(x) = 2x+1$$

Ex-② $X = (0, \frac{1}{4}]$ $T: X \rightarrow X$ $Tx = x^2$

$$|Tx - Ty| = |x^2 - y^2|$$

$$= |x-y||x+y|$$

$$< \frac{1}{2} |x-y|$$

$$0 < x < \frac{1}{4}$$

$$0 < y < \frac{1}{4}$$

This is a contraction mapping $\alpha = \frac{1}{2}$ having no fixed pt.

Ex-3

$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$

fixed points = all the pts of x -axis.

Ex-4 :- $X = [0, 1]$ defined $T: [0, 1] \rightarrow [0, 1]$ by $T(x) = \cos x$.

Is T contraction on X . If so then what is the fixed pt of T .

$$|T(x) - T(y)| = |\cos x - \cos y| = \left| 2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \right|$$

$$\leq 2 \left| \frac{x-y}{2} \right|$$

$$= |x-y|$$

$$\alpha = 1.$$

$\therefore T$ is not contraction.

Application of Banach ~~sp~~ fixed pt theorem in linear algebra:-

Let $y_i = \sum_{j=1}^n a_{ij} x_j + b_i \quad i=1, 2, \dots, n$ can be thought of as a mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $y = Tx = ax + b$ where $x = (x_1, \dots, x_n)^T$ $y = (y_1, \dots, y_n)^T$ $b = (b_1, b_2, \dots, b_n)^T$. We now determine sufficient conditions on T to be contraction mapping under different choices of metric on $\mathbb{R}^n$.

Case - I: Under the metric defined on $\mathbb{R}^n$ by

$$d_2(x, y) = \left[\sum_{i=1}^n (x_i - y_i)^2 \right]^{\frac{1}{2}} \quad x, y \in \mathbb{R}^n$$

$$\text{Now, } d_2^2(y, \tilde{y}) = \sum_{i=1}^n (y_i - \tilde{y}_i)^2 = \sum_{i=1}^n \left[\sum_j a_{ij} (x_i - \tilde{x}_i) \right]^2$$

$$\leq \sum_{i=1}^n \left[\left(\sum_j a_{ij}^2 \right) \left(\sum_k (x_i - \tilde{x}_i)^2 \right) \right]$$

$$d_2^2(y, \tilde{y}) \leq \left(\sum_{j,j} a_{ij}^2 \right) d_2^2(x, \tilde{x})$$

Thus if $\sum_i \sum_j a_{ij}^2 < 1$ then T is a contraction mapping on $(\mathbb{R}^n, d_2)$

ℓ^p space

$\ell^p = \{x = (x_1, x_2, \dots) : \|x\|_p = \left(\sum_{k=1}^{\infty} \|x_k\|^p\right)^{1/p} < \infty\}$
 $x_k \in \mathbb{R} \text{ or } \mathbb{C}$
space of absolutely summable sequence over $\mathbb{R}$ or $\mathbb{C}$.

Equivalent norm

$$\|\cdot\|_1, \|\cdot\|_2$$

$\exists m, M$

$$m\|x\|_1 \leq \|x\|_2 \leq M\|x\|_1 \quad \forall x \in \ell^2$$

Application in the theory of Differential eqnⁿ

Picard's theorem :-

Let S be the rectangle where $S = \{(x, y) \in \mathbb{R}^2 \mid |x - x_0| \leq a, |y - y_0| \leq b\}$

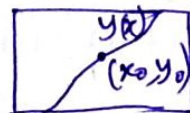
or $S = [x_0 - a, x_0 + a] \times [y_0 - b, y_0 + b]$.

Suppose that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous on S and $\exists$ a constant L such that $|f(x, y_1) - f(x, y_2)| \leq L|y_1 - y_2|$
 $\forall (x, y_i) \in S, y_i = 1, 2, \dots$

let $\alpha = \min \left\{ a, \frac{b}{M}, \frac{1}{L} \right\}$ where $|f(x, y)| < M \quad \forall (x, y) \in S$.

Then the initial value problem $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$ has a unique solⁿ in the interval $[x_0 - \alpha, x_0 + \alpha]$.

proof:-



Let $I = [x_0 - \alpha, x_0 + \alpha]$ and $J = [y_0 - b, y_0 + b]$

Let $C(I, J)$ be the space of continuous funcⁿ from I to J with the metric $d(\phi_1, \phi_2) = \sup_{t \in I} |\phi_1(t) - \phi_2(t)|$

[Note:- $e(x, y) = \{f: x \rightarrow y \mid f \text{ is continuous}\}$
 $d(f, g) = \sup_{x \in X} [d_Y(f(x), g(x))]$
 $(C(X, Y), d)$ is complete if (Y, d_Y) is complete.]

let $Y = C(I, \mathbb{R})$

clearly Y is a subspace of $X = C(I, \mathbb{R})$ where the metric on X is defined as $\rho(f, g) = \sup_{t \in I} |f(t) - g(t)|$

To ensure that Y is a complete metricspace we need only to show that Y is closed in X .

Let $\phi \in \bar{Y}$

$\Rightarrow \exists$ a sequence $\{\phi_n\}_{n \geq 1}$ in Y such that $\{\phi_n\} \rightarrow \phi$ in X . (as X is complete).

Now for all $t \in I$, we have $|\phi_n(t) - \phi(t)| \leq \rho(\phi_n, \phi) \rightarrow 0$ as $n \rightarrow \infty$ i.e. $\lim_{n \rightarrow \infty} \phi_n(t) = \phi(t) \quad \forall t \in I$.

clearly $y_0 - b \leq \phi_n(t) \leq y_0 + b \quad \forall t \in I$.

$$\Rightarrow y_0 - b \leq \lim_{n \rightarrow \infty} \phi_n(t) \leq y_0 + b \quad \forall t \in I.$$

$$\Rightarrow y_0 - b \leq \phi(t) \leq y_0 + b \quad \forall t \in I.$$

$$\Rightarrow \phi(t) \in Y \quad \forall t \in I.$$

Hence Y is closed in X . i.e. (Y, d_Y) is a complete metricspace.

Define a mapping T on Y by $T\psi(x) = y_0 + \int_{x_0}^x f(t, \psi(t)) dt$

Then $T\psi$ is differentiable then

$$(T\psi)'(x) = f(x, \psi(x))$$

$\Rightarrow T\psi$ is continuous

$$\begin{aligned} \text{Moreover } |T\psi(x) - y_0| &= \left| \int_{x_0}^x f(t, \psi(t)) dt \right| \\ &\leq \int_{x_0}^x |f(t, \psi(t))| dt \\ &\leq M |x - x_0| \leq M a < b \end{aligned}$$

$$\Rightarrow T\psi \in Y$$

By the Lipschitz condition $dy(T\psi_1, T\psi_2) = \sup_{t \in I} \left| \int_{x_0}^x f(t, \psi_1(t)) dt - \int_{x_0}^x f(t, \psi_2(t)) dt \right|$

$$\leq \sup_{t \in I} \left| \int_{x_0}^x |f(t, \psi_1(t)) - f(t, \psi_2(t))| dt \right|$$

$$\leq \sup_{t \in I} \left| \int_{x_0}^x [L dy(\psi_1, \psi_2)] dt \right|$$

$$\leq L dy(\psi_1, \psi_2) |x - x_0|$$

$$\leq \alpha L \cdot dy(\psi_1, \psi_2)$$

So, $T: Y \rightarrow Y$ is contraction mapping since $\alpha L < 1$.

$\Rightarrow T$ has a unique fixed point say $y \in Y = C(I, J)$

satisfying $Ty = y \Rightarrow y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$

clearly $y(x_0) = y_0$

since the ~~to~~ RHS is differentiable ~~y is diff~~ $y(x)$ is differentiable and $y'(x) = f(x, y(x))$ i.e. $y(x)$ is the soln of

$$\frac{dy}{dx} = f(x, y(x)) \quad \forall x \in I \text{ with } y(x_0) = y_0.$$

Note:- Any soln of this ode with this initial condition is a fixed pt. of T , $T y(x) = y_0 + \int_{x_0}^x f(t, y(t)) dt$

$$= y_0 + \int_{x_0}^x \frac{dy}{dt} dt$$

$$= y(x) \quad \forall x \in I.$$

$$\boxed{Ty = y}$$

① $\frac{dy}{dx} = x|y|$, $y(0) = 1$

check that soln is unique??

② $\left| \frac{dy}{dx} \right| + |y| = 0$, $y(0) = 1$